Unit-IV

Introduction To Hyperledger

Hyperledger **provides the platform** to **create personalized blockchain services according to the need of business work**. Unlike other platforms for developing blockchain-based software, Hyperledger has the advantage of creating a secured and personalized blockchain network.

- It is created **to support** the development of **blockchain-based distributed ledgers**.
- It includes a **variety of enterprise-ready permissioned** blockchain platforms.
- It is a global collaboration for developing high-performance and reliable blockchain and distributed ledger-based technology frameworks.

Example: Consider a situation **when person X wants to buy medicine from person Y,** who was a doctor living in another country.

- As **medicine requirement** is one person's private needs, they need **to maintain the data confidentially.**
- But Dr. Y is **selling medicine in the network** to so many people, in the case of the public blockchain, **every transaction will get updated** in the network to **all the peers.**
- That's where hyperledger finds its significance. **In the hyperledger**, the parties are **directly connected and the concerned people's ledger will be updated**.
- Hence providing privacy and confidentiality.

Why Do We Need Hyperledger?

Below are some of the reasons stating the need for a hyperledger project:

- To **enhance the efficiency, performance, and transactions** of various business processes.
- It provides the **necessary infrastructure and standards for developing various blockchain-based systems** and applications for industrial use.
- It gets **rid of the complex nature of contractual agreements, as the legal issues** are taken care of.
- Hyperledger offers the **physical separation of sensitive data**.
- It decreases the need for verification and enhances trust, thus **optimizing network performance and scalability.**

<u>Hyperledger Technology Layers</u>

Hyperledger uses the following key business components:

1. **Consensus layer:** It **takes care of creating an agreement on the order** and confirming the correctness of the set of transactions that constitute a block.
2. **Smart layer:** This layer is **responsible** for **processing transaction** requests and **authorizing valid transactions**.
3. **Communication layer:** It takes care of **peer-to-peer message** transport.
4. **Identity management services:** these are important for **establishing trust** on the blockchain.
5. **API:** It **enables external applications and clients to interface with the blockchain**.

How Does Hyperledger Work?

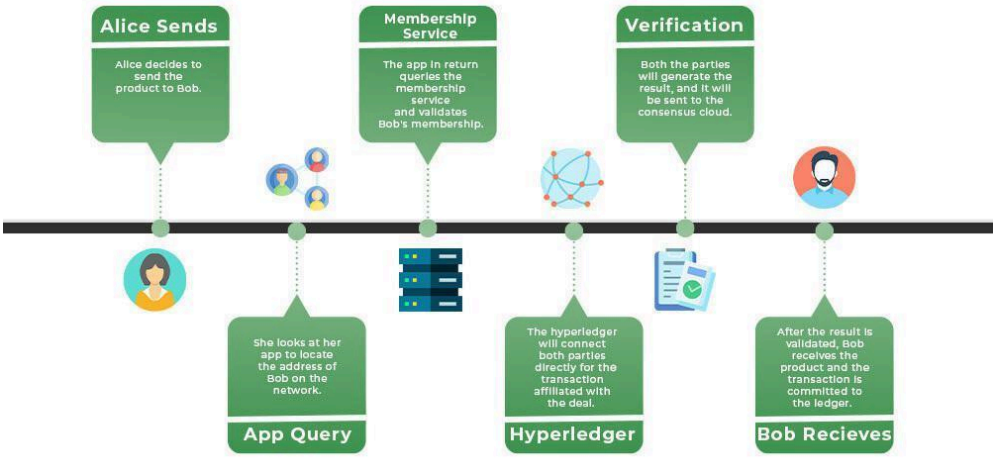
Hyperledger works in a way that a requirement for the contract can be initiated through an application.

- The **membership service involved** in the network **validates the contract**.
- The **concerned two-peer has to produce a result** and then **sent it to the consensus cloud.**
- The **generated result from both the peer has to be the same** in order to **validate the contract.**
- **Once it is validated**, then the **transaction will happen** between the affiliated peers and their **ledger will be updated**.

  When a business requires **confidentiality** and a private network for its transaction to happen without doing that in a single network, a **hyperledger** paves the way.

Let's understand the concept with the help of an example.

- Suppose Alice decides to **send the product** to Bob on a hyperledger-based network.
- She looks at her app to **locate her address** of Bob on the network.
- The app in return **queries the membership service** and validates Bob's membership.
- The **hyperledger will connect both parties** directly for the transaction affiliated with the deal.
- **Both parties will generate the result**, which has to be the **same** for them to get validated.
- The **result will be sent to the consensus** cloud to be **ordered and verified**.
- **Once the result is validated**, Bob receives the product and the transaction is committed to the ledger.



<u>What is Distributed Ledger Technology (DLT)?</u>

Distributed Ledger Technology (DLT) is centered around an encoded and distributed database where **records regarding transactions are stored**. A distributed ledger is a **database** that is **spread across various computers, nodes, institutions, or countries** accessible by multiple people around the globe.

Types of Distributed Ledger Technology

The Distributed Ledgers can be categorized into three categories:

1. **Permissioned DLT: Nodes have to take permission from a central authority** to access or make any changes in the network. Mostly these types of permissions include identity verification.
2. **Permissionless DLT: There is no central authority to validate transactions**, rather existing nodes are collectively responsible for validating the transactions. Various consensus mechanisms are used to validate transactions based on predefined algorithms. In the case of bitcoin proof of work consensus mechanism is used.
3. **Hybrid DLT:** It is **combined** with both permissionless and permissioned DLTs and can benefit from both of them.

Distributed Ledger Technology Challenges

- Lack of clarity on the terminology and perceived immaturity of the technology.
- Perceived risks in early adoption and likely disruption to existing industry practices.
- Insufficient evidence on business gains and wider economic impact.
- Lack of clarity on how the technology is/would be governed.
- Uncertainty around regulation.
- Multiple non-interoperable implementations and resulting fragmentation.
- Maintaining security and privacy of data.
- Ensuring integrity of data and strong encryption.
- Energy-intensive nature of the technology.
- Lack of clarity regarding smart contracts and how to implement them through DLT

What is Hyperledger Fabric ?

Hyperledger Fabric is designed for use in **enterprise-level applications**, and it is characterized by its modular architecture, permissioned network, and smart contract functionality, known as "chaincode".

- The platform provides a **high degree of security, privacy, and scalability**, and it supports the development of custom blockchain solutions for various use cases across industries such as **finance, supply chain, and healthcare.**
- Hyperledger Fabric **operates as a network of nodes,** where **each node performs a specific function,** such as **validating transactions, maintaining the ledger, and executing chaincode.**
- Transactions are validated and ordered by a **consensus mechanism**, which ensures the integrity and consistency of the ledger.

How does Hyperledger Fabric Work?

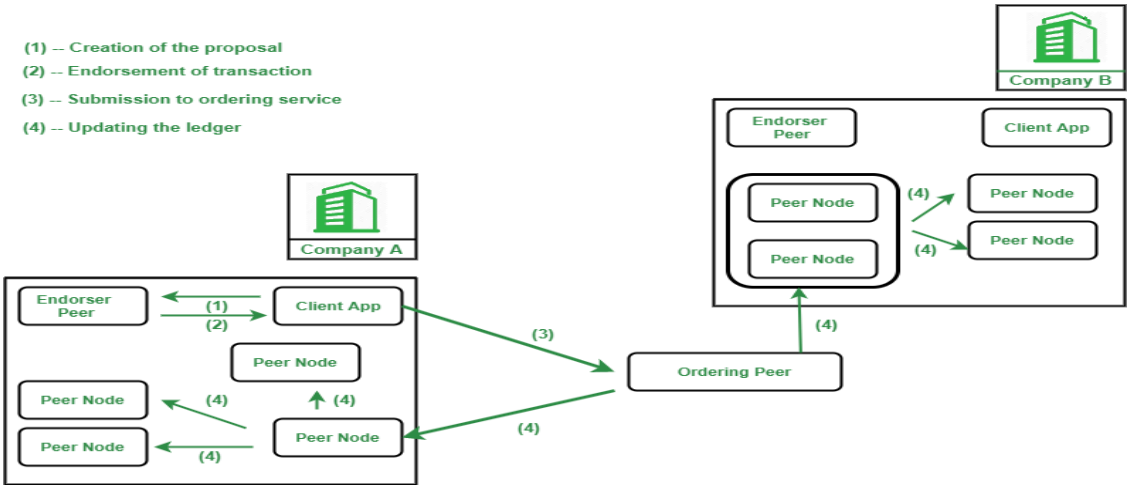
Components:

- Hyperledger fabric is an **enterprise-level permission blockchain network**. It is made up of **various unique organizations or members that interact with each other to serve a specific purpose.** For example, these organizations can be a **bank, financial institution, or a supply chain network**. Each organization is identified and they have a fabric certificate authority. These organizations are called members.
- Each member of the fabric can set up one or more authorized peers to participate in the network using the fabric certificate authority. All of these peers must be authorized properly.
- There is a client-side application connected to the network written with the **software development kit (SDK)** of any particular programming language.

Workflow:

For each and every transaction in the fabric, the following steps are followed-

1. **Creation of the proposal:** Imagine a deal between a smartphone manufacturer company and a smartphone dealership. **The transaction begins when a member organization proposes or invokes a transaction request with the help of the client application or portal.** Then the client application sends the proposal to peers in each organization for endorsement.
2. **Endorsement of the transaction:** After the proposal reaches the endorser peers (peers in each organization for endorsement of a proposal) the **peer checks the fabric certificate authority** of the requesting member and other **details that are needed to authenticate the transaction**. Then it **executes the chain code** (a piece of code that is written in one of the supported languages such as **Go or Java**) and **returns a response.** This response indicates the approval or rejection of the following transaction. The response is carried out to the client.
3. **Submission to ordering service:** After **receiving the endorsement output**, the approved **transactions are sent to the ordering service by the client-side application**. The peer responsible for the ordering service includes the transaction into a specific block and sends it to the peer nodes of different members of the network.
4. **Updating the ledger: After receiving this block the peer nodes of such organizations update their local ledger with this block**. Hence the new transactions are now committed.



Hyperledger Fabric Consensus Algorithm

Hyperledger Fabric uses a consensus algorithm to **achieve agreement among the participants in a network** on the contents of the shared ledger. The consensus algorithm in Hyperledger **Fabric is pluggable**, which means that it **can be replaced with a different algorithm as needed.**

The most commonly used consensus algorithms in Hyperledger Fabric are:

- **Practical Byzantine Fault Tolerance (PBFT):** PBFT is a consensus algorithm that **provides fault tolerance** and **reliability in a network**. It is well-suited for networks with a **limited number of participants** who are **trusted and well-known.**
- **RAFT**: RAFT is a consensus algorithm that is used to **maintain a consistent state across multiple nodes**. It is well-suited for networks where the **participants are unknown and potentially untrusted.**
- **Solo**: Solo is a consensus algorithm that is used for **testing purposes in a single-node network**. It is not suitable for production use.

Benefits Of Hyperledger Fabric

1. Open Source: Hyperledger fabric is an open-source blockchain framework hosted by the Linux foundation. It has an active community of developers The code is designed to be publicly accessible. Anyone in the community can see, modify, and distribute the code as they see fit.

2. Private and Confidential: In a **public blockchain network** each and every node in the network is receiving a copy of the whole ledger. Thus keeping privacy becomes a much bigger concern as everything is open to everyone. In addition to this one, the identities of all the participating members are not known and authenticated. Anyone can participate as it is a public blockchain. But in the case of **Hyperledger fabric**, the identities of all participating members are authenticated. And the ledger is only exposed to the authenticated members. This benefit is the most useful in industry-level cases, like **banking, insurance**, etc where customer data should be kept private.

3. Access Control: In the Hyperledger fabric, there is a **virtual blockchain network on top of the physical blockchain network**. It has its own access rules. It employs its own mechanism for transaction ordering and provides an additional layer of access control**. It is especially useful when members want to limit the exposure of data and make it private.** Such that it **can be viewed by the related parties only**. As an example when two competitors are on the same network. **The fabric also offers private data collection and accessibility, where one competitor can control the access to its own data such that the data do not get exposed to the other competitor.**

4. Chaincode Functionality: It includes a container technology to **host smart contracts called chain code that defines the business rules of the system**. And it's designed to **support various pluggable components and to accommodate the complexity that exists across the entire economy.** This is useful for some of the specific types of transactions like asset ownership change.

5. Performance: As the Hyperledger fabric is a private blockchain network, There **is no need to validate the transactions** on this network so the **transaction speed is faster**, resulting in a better performance.

Limitation of Hyperledger Fabric

Hyperledger Fabric is a robust and flexible platform for developing blockchain applications, but like any technology, it has certain limitations:

1. **Scalability:** Hyperledger Fabric is **designed for permissioned networks,** where the participants are known and trusted, which can **limit its scalability** for large-scale public networks.
2. **Performance:** The performance of Hyperledger Fabric can be impacted by factors such as network size, network configuration, and the complexity of chaincode, which can **limit its ability** to handle high volumes of transactions.
3. **Complexity:** Setting up and configuring a Hyperledger Fabric network can be **complex, requiring a deep understanding** of the technology and its components.
4. **Compatibility:** Hyperledger Fabric is designed to be used with **specific programming languages, such as Go and JavaScript, which can limit its compatibility** with other technologies and programming languages.
5. **Cost:** Running a Hyperledger Fabric network requires infrastructure and resources, which can **add costs to the deployment and operation** of blockchain applications.
6. **Interoperability:** Hyperledger Fabric is **designed to be used within a single network**, and its interoperability with other blockchain platforms is limited.

What is Hyperledger Composer?

Hyperledger Composer is a **framework for building blockchain business networks**. It provides a high level of abstraction for designing your network so you can **focus on the specific application and get it running quickly**. **There is flexibility to use the toolkit provided or share them with other players in your own industry space so that you don't have to reinvent the wheel when you start working on your own applications from scratch.**

- Carrying the burden of complexity in business transactions inevitably necessitates a large number of contract codes in order to make sure that certain agreements are fulfilled.
- The need for maintaining these contracts can ultimately lead to a lack of consistency between codes as well as lost time and resources when mistakes happen during the process.
- To solve this problem, it's important to use technology like Hyperledger Composer which facilitates shared cross-project code management for all stakeholders involved in development and deployment rather than relying on a single entity's interpretation or attempts at execution.

Key Points:

- Hyperledger Composer is an **open-source platform** that supports smart contract development.
- The platform focuses on allowing users to create business networks by means of **making use of pre-built components**.
- It's designed for supporting blockchain applications in order **to make them compatible with other Hyperledger tools.**
- It's developing and **maintainable by different stakeholders at the same time, not just a single entity.**
- The platform's development team aims to make the platform readily available to developers, organizations, businesses, and associations which are looking to utilize

blockchain technologies like Hyperledger Composer in order to conduct business operations.

- The platform focuses on cross-project code management for all stakeholders involved in the development and deployment of their projects.
- Hyperledger Composer is a modular architecture that facilitates the creation of components that can be incorporated and reused in order to reduce complexity.

In order to reduce complexity while supporting secure transactions in a digital environment, the developers of Hyperledger Composer focus on the creation of components that can be incorporated and reused in order to reduce complexity while ensuring secure transactions. It's also designed to support blockchain applications and is capable of making them compatible with other Hyperledger tools. It's developed and maintained by different stakeholders at the same time, not just a single entity. It's an open-source tool that facilitates the creation of business networks through its modular architecture.

Features Of Hyperledger Composer

- Hyperledger composer is a modular toolkit with pre-built components to build blockchain business networks. With its help, you can build blockchain business networks in days and not weeks.
- You can create a smart contract using composer and get it running in only two days.
- It also supports plug-in features that improve debugging experience by enabling users to troubleshoot network-related issues effectively. It also lets developers assess the network communication flows easily with network packet traces and decoding of the payloads generated by different applications connected to the distributed ledger system.
- One can use one of these sample plug-ins to troubleshoot the system runtime environment of your choice on Mac, Linux, or Windows systems.

Hyperledger Composer Architecture

Developers of the Hyperledger Composer platform utilize Apache Cassandra as a tool for providing access to a set of services that form the underlying architecture of this platform.

- These services include the ledger, storage interface, and event service.
- In order to interact with these services and make use of the Hyperledger Composer platform, developers can make use of the application called "Composer Playground" which allows them to get used to the workflow and perform certain tasks like creating contracts or running different transactions through an IDE (Integrated Development Environment).

Hyper Ledger Composer Working

Hyperledger composer is a business toolkit that is used for developing blockchain business networks. One can create a network using composer and transact with digital assets such as smart properties, contracts, value tokens, and currencies by using their unique digital identities.

- It provides the freedom to create your own applications with it because of its modular architecture.
- One can start from scratch, import an existing Ethereum Solidity smart contract or use one of its sample plug-ins like Hyperledger Geth to troubleshoot the system runtime environment on Mac, Linux, or Windows systems.
- Hyperledger Composer is a platform that's geared towards promoting consistency in business transactions through the use of a modular architecture that facilitates the creation of components that can be incorporated and reused in order to reduce complexity.
- It's an open-source tool that supports smart contract development and focuses on allowing users to create business networks by means of making use of pre-built components. It's also focused on supporting blockchain applications that are compatible with other Hyperledger tools.

The development of Hyperledger Composer started in August 2015 as a result of the blockchain-based voting platform having been released by the Ethereum Foundation. According to the project's website, a Hyperledger Composer is a tool that was designed to reduce complexity while supporting secure transactions in a digital environment. It can be used for maintaining complex business networks and is designed to manage shared code across organizations. The platform's development team aims to make the platform readily available to developers, organizations, businesses, and associations which are looking to utilize blockchain technologies like Hyperledger Composer in order to conduct business operations.

Benefits Of Hyperledger Composer

- Hyperledger composer helps in building various business applications and allows you to experiment with the technology at a low cost.
- It lets you quickly modify the model from one application to another.
- It is an open-source tool which means that you can get it for free. The code is hosted on GitHub and anyone can contribute by adding or modifying features to this framework.
- The best part is that it can be used by developers as well as non-technical users in organizations, regardless of their level of proficiency with blockchain technology.

Conclusion

- While Hyperledger Composer is a platform that supports smart contract development, it's essentially an open-source tool that can be used by developers to create and maintain their own applications. Its modular architecture allows users to develop components that can be incorporated and reused in order to reduce complexity while ensuring secure transactions. It's developed and maintained by different stakeholders at the same time, not just a single entity. It facilitates shared cross-project code management for all stakeholders involved in the development and deployment of contracts or projects. Thus, developers can make use of this platform in order to create a business network using pre-built components or to make blockchain technologies compatible with other Hyperledger tools.